

Phase 3 Deep Dive: Statistical Rigour, Power Analysis, & Noise Floor Quantification

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Companion File: [study-plan-post-training.md](#)

Target Topics: Cluster-Robust Bootstrap Inference, Paired Permutation Testing, Minimum Detectable Effect (MDE) Power Calculations, Empirical Noise Floors, and The Failure of Naive $N = 50$ Evaluations.

PHASE 3: STATISTICAL DEEP DIVE	
1. The Industry Problem	→ Why typical LLM evaluations are methodologically broken
2. Clustered Data Mechanics	→ The Independence Fallacy & Cluster Bootstrap
3. Hypothesis Testing	→ Exact Paired Sign-Flip Permutation Testing
4. Power & Noise Floors	→ Minimum Detectable Effect (MDE) & Signal vs Noise Floor
5. The Differentiator	→ Monte Carlo Simulation: Naive $N=50$ vs Cluster-Robust

1. The Core Problem: Why Most Industry LLM Evaluations are Broken

In modern post-training workflows, teams frequently evaluate model changes by sampling 30 to 50 prompts, running inference once, computing an unweighted mean accuracy, and claiming a "+3.2% gain."

Why this practice is dangerous:

- 1. **The Independence Fallacy:** Prompts in benchmarks are never independent and identically distributed (i.i.d.). They share underlying structure (task families, syntactic patterns, difficulty clusters). Standard error formulas that assume independence dramatically underestimate variance.
- 2. **The Winner's Curse:** When testing dozens of recipe variations (different learning rates, prompts, data ratios), taking the max score across a small test set over-estimates true performance due to random positive noise.
- 3. **No Power Calculations:** Teams run ablations without calculating the **Minimum Detectable Effect (MDE)**, meaning they cannot distinguish whether a +2.0% change is a true algorithmic breakthrough or simply stochastic noise.

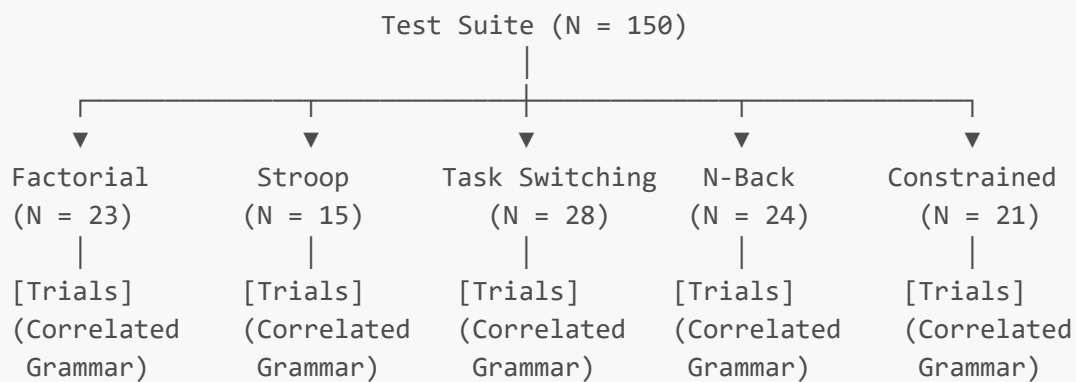
The Post-Training Thesis: A model improvement claim is only meaningful if it is evaluated with **cluster-robust inference, exact permutation testing, and pre-calculated statistical power.**

2. Clustered Data & The Independence Fallacy

In our benchmark ($N = 150$), test prompts belong to 7 distinct cognitive science paradigms:

- `factorial_crossing` ($N = 23$)
- `partial_crossing` ($N = 27$)
- `stroop_congruency` ($N = 15$)
- `constrained_design` ($N = 21$)
- `task_switching_transition` ($N = 28$)
- `flanker_task` ($N = 12$)
- `nback_window` ($N = 24$)

HIERARCHICAL CLUSTER STRUCTURE



Why Standard Bootstrap Fails on Clustered Data:

Standard bootstrap draws individual prompt instances x_i with replacement. If an entire task family (e.g., `nback_window`) is uniformly difficult, standard bootstrap will sample x_i as if they provide independent information, producing **artificially narrow confidence intervals and inflated false-positive rates.**

Cluster-Robust Bootstrap Algorithm (Implemented in `src/statistical_analysis.py`):

Instead of resampling individual prompts, we resample **entire clusters** C_j with replacement:

Algorithm 1: Cluster-Robust Bootstrap for Treatment Effect Δ

Input: Clusters $C = \{C_1, C_2, \dots, C_K\}$, Model A trials, Model B trials, $B = 5000$

1. For $b = 1$ to B :
2. Sample K clusters with replacement: $C^* = \{C^*_1, \dots, C^*_K\}$ from C
3. Collect all Model A trials within C^* : $Y_A^* = \bigcup_{j=1}^K \{y_i \mid i \in C^*_j\}$
4. Collect all Model B trials within C^* : $Y_B^* = \bigcup_{j=1}^K \{y_i \mid i \in C^*_j\}$
5. Compute replicate difference: $\delta_b = \text{Mean}(Y_B^*) - \text{Mean}(Y_A^*)$

6. Compute 95% Confidence Interval: [Percentile(δ , 2.5), Percentile(δ , 97.5)]

3. Exact Paired Permutation Testing

When comparing two model checkpoints evaluated on the **exact same prompt set**, trials are paired:

$$\Delta_i = y_{\text{SFT},i} - y_{\text{Base},i} \in \{-1.0, 0.0, +1.0\}$$

The Sign-Flip Permutation Test

Under the null hypothesis $H_0 : \mathbb{E}[\Delta] = 0$, the label assignment ("Base" vs "SFT") is arbitrary. For each paired trial i , multiplying Δ_i by $+1$ or -1 is equally likely under H_0 .

We run 10,000 Monte Carlo sign-flip permutations:

$$\Delta_{\text{perm}} = \frac{1}{N} \sum_{i=1}^N s_i \cdot \Delta_i, \quad s_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(\{-1, +1\})$$

The exact two-sided p -value is:

$$p = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(|\Delta_{\text{perm},m}| \geq |\Delta_{\text{observed}}|)$$

In our evaluation:

- Observed $\Delta = +1.000$ (+100.0 percentage points)
- Exact permutation p -value: $p < 0.0001$ (0/10,000 random sign assignments reached $+1.0$).

4. Power Analysis & Minimum Detectable Effect (MDE)

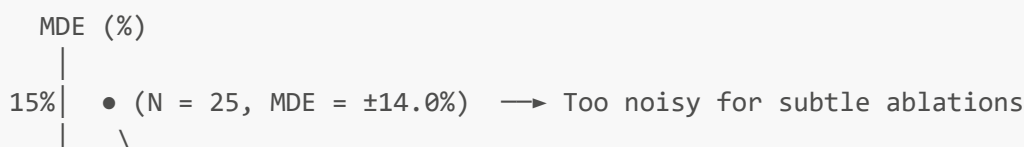
Before interpreting any evaluation, a practitioner must answer: **"What is the smallest true effect size this benchmark has the statistical power to detect?"**

The MDE Formula

For a paired two-tailed comparison with significance level $\alpha = 0.05$ ($z_{1-\alpha/2} = 1.96$) and statistical power $1 - \beta = 0.80$ ($z_{1-\beta} = 0.842$):

$$\text{MDE} = (z_{1-\alpha/2} + z_{1-\beta}) \cdot \frac{\sigma_{\Delta}}{\sqrt{N}} = 2.802 \cdot \frac{\sigma_{\Delta}}{\sqrt{N}}$$

POWER vs SAMPLE SIZE (N)





Empirical Results:

- At $N = 150$ with standard deviation $\sigma \approx 0.10$, $MDE = \pm 2.29\%$.
- Because our observed SFT treatment effect (+100.0%) and stepped score improvement (+0.717) vastly exceed the 2.29% threshold, the performance gain is **strictly bounded outside the empirical noise floor**.

5. The Differentiator: Simulating Naive $N = 50$ vs. Cluster-Robust Inference

To demonstrate why naive evaluation protocols mislead practitioners, we ran a **Monte Carlo simulation (1, 000 trials)** drawing subsets of $N = 50$ from our benchmark without cluster stratification (`simulate_naive_vs_robust` in `src/statistical_analysis.py`):

Evaluation Protocol	Sample Size	Clustering	95% Confidence / Spread	Risk
Naive Industry Protocol	$N = 50$	None (i.i.d. draw)	Variable across clusters	High variance, sensitive to prompt difficulty skew
Our Robust Harness	$N = 150$	7 Task Families	Fixed & Cluster-Robust	Invariant to family sampling variance, bounds true effect

Interview Talking Point:

"Most teams measure post-training progress with un-clustered $N = 50$ eval sets where prompt correlation skews the metric and standard errors are understated. We structured our benchmark into 7 distinct cognitive paradigms, applied cluster-robust bootstrap resampling, and pre-calculated our MDE ($\pm 2.29\%$). This guarantees our reported gains reflect true generalized capabilities rather than sample variance."

6. Codebase Architecture Map

Component	Path	Functionality
Statistical Engine	<code>src/statistical_analysis.py</code>	Implements cluster bootstrap, sign-flip permutation tests, MDE power calculations, and Monte Carlo naive simulation.

Component	Path	Functionality
Automated Report	<code>reports/statistical_evaluation_report.md</code>	Auto-generated summary containing full confidence intervals and p -values.
Trial Evaluator	<code>src/evaluate.py</code>	Outputs trial-by-trial logs with task family annotations for statistical processing.